

Rebates in Supply Chains

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Abstract

Textbook publishers frequently agree to buy unsold textbooks back from campus bookstores, presumably in an effort to increase order quantities and sales. We consider a model in which publishers jointly set wholesale prices and rebate policies, and bookstores solve an associated Newsvendor problem. We solve for optimal wholesale prices, rebates, and order quantities, and identify two key insights. First, when the publisher is a monopolist, it is optimal to set wholesale prices such that bookstores make almost no profit from each sale, but to offer a full rebate. Second, increased competition causes the publisher to lower wholesale prices and offer a less generous rebate policy – in the extreme case of a perfectly competitive market, wholesale prices are equal to the marginal cost of provision and no rebate is offered.

1 Model

To highlight the generality of the model, in what follows we refer to the publisher as a “supplier”, and the bookstore as a “retailer.”

A supplier can provide a retailer with items to sell. The retail price of the item is fixed to be p , and at that price, customer demand for the item (from the retailer) is distributed according to a known distribution with cdf F .¹ The marginal cost to the supplier of providing an additional book is c_m (this may account for manufacturing and/or shipping costs), and the retailer incurs a holding cost of c_h for each item purchased. We assume that $p > c_m + c_h$ and define the *margin* of the product to be $M = (p - c_m - c_h)/p$ (this is the percentage of the sale price that serves as profit for either the supplier or the retailer). A reader looking for the “simplest” version of this model can assume $p = 1$ and $c_h = 0$, as this is sufficient to reveal all major insights. The game is as follows:

1. The supplier chooses a wholesale price w , and a rebate r that will be offered for each unsold item.
2. The retailer chooses an order quantity q .
3. Demand $d \sim F$ is realized, and any unsold books are returned to the supplier.

¹For simplicity of notation, in what follows we assume that F is continuous and strictly increasing on some interval $[\underline{d}, \bar{d}] \subseteq [0, \infty)$, with $F(\underline{d}) = 0$ and $F(\bar{d}) = 1$, so that F^{-1} is well-defined.

The utilities of the supplier and retailer given wholesale price w , rebate r , order quantity q and demand d are respectively

$$\begin{aligned} U_S(w, r, q, d) &= (w - c_m)q - r \max(q - d, 0) \\ U_R(w, r, q, d) &= p \min(q, d) + r \max(q - d, 0) - q(w + c_h) \\ &= q(p - w - c_h) - (p - r) \max(q - d, 0). \end{aligned}$$

Both the supplier and retailer are risk-neutral. In what follows, we make use of the following identity.

$$\mathbb{E}_{d \sim F}[\max(q - d, 0)] = \int_0^q F(x) dx \quad (1)$$

We assume that whenever the retailer is indifferent between different order quantities, the supplier chooses which of these quantities will be ordered by the retailer.²

2 Monopolist Supplier

As is typical with Stackelberg games, we solve by backwards induction. Given wholesale price w and rebate $r \leq w + c_h$,³ a retailer that orders quantity q earns expected reward of

$$\mathbb{E}_{d \sim F}[U_R(w, r, q, d)] = q(p - w - c_h) - (p - r) \int_0^q F(x) dx \quad (2)$$

If $p > r$, this has a unique local maximum, given by⁴

$$q(w, r) = F^{-1} \left(\frac{p - c_h - w}{p - r} \right). \quad (3)$$

The supplier chooses r and w to solve the following:

$$\mathbb{E}_{d \sim F}[U_S(w, r, q(w, r), d)] = (w - c_m)q(w, r) - r \int_0^{q(w, r)} F(x) dx.$$

To solve the supplier's problem, we note that for fixed $r < p$, there is a one-to-one correspondence between wholesale prices w and order quantities $q(w, r)$. In particular, we have

$$w = p - c_h - (p - r)F(q(w, r)).$$

²This is analogous to letting the mechanism designer select their preferred equilibrium in auction theory. It ensures that the supremum of the supplier's expected profit is attainable. If instead we assumed that the retailer chose among optimal order quantities antagonistically, then the supplier could get arbitrarily close to this supremum by using small deviations from the policies stated in this paper.

³If $r > w + c_h$, then the retailer prefers to order as many items as possible, as unsold books are profitable.

⁴In the language of the standard Newsvendor problem, the retailer faces overage cost $c_o = w + c_h - r$ and underage cost $c_u = p - w - c_h$, and chooses an order quantity q such that $F(q) = \frac{c_u}{c_u + c_o}$. If $p = w + c_h = r$, then c_u and c_o are undefined, and the retailer is indifferent between all order quantities.

This allows us to perform a change of variables: instead of choosing a wholesale price and rebate, we imagine that the supplier chooses a rebate and a quantity. In this case, the optimization problem becomes

$$\max_{r,q} (p - c_h - (p - r)F(q) - c_m)q - r \int_0^q F(x)dx. \quad (4)$$

Rearranging yields

$$\max_{r,q} (p(1 - F(q)) - c_h - c_m)q + r \left(qF(q) - \int_0^q F(x)dx \right).$$

Note that $qF(q) - \int_0^q F(x)dx > 0$ for any q such that $F(q) > 0$, because F is increasing. It follows that for any q , the supplier's profit is increasing in r . Thus, the supplier wishes to select $r = p$ (and $w = p - c_h$). Substituting this into (4) shows that the optimal order quantity is

$$q^* = F^{-1}(M). \quad (5)$$

We have just established the following result.

Theorem 1. *A monopolist supplier chooses wholesale price $w = p - c_h$ so that the retailer makes no profit on each sale, but offers a full rebate $r = p$ for all unsold items. In the supplier-optimal equilibrium, the retailer orders $q = q^* = F^{-1}(M)$, and makes no profit. The supplier's expected profit is*

$$\Pi^* \triangleq p \left(q^* F(q^*) - \int_0^{q^*} F(x)dx \right) \quad (6)$$

Remark 1. *Theorem 1 states that a monopolist publisher will choose a wholesale price such that the bookstore makes essentially no profit, and offer a full rebate to encourage more sales. This sounds descriptively similar to common practice in the textbook industry.*

Remark 2. *It is helpful to contrast this outcome with the outcome where no rebate is offered. In that case, (4) implies that the supplier chooses q to maximize*

$$\max_q q(p(1 - F(q)) - c_h - c_m).$$

Note that this is the familiar pricing problem of a monopolist with marginal cost of production c_m facing demand distribution $Q(w) = 1 - F^{-1}\left(\frac{w+c_h}{p}\right)$.

It can be shown that this results in a lower order quantity, and an outcome in which the retailer earns positive expected profits. In addition, if the model were extended to allow multiple retailers with different demand distributions, then the supplier's optimal wholesale price (when r is constrained to be zero) is different for each retailer. This may not be practical. By contrast, introducing rebates allows the supplier to offer a "fair" policy (the same wholesale prices and rebates for all retailers, regardless of the demand distribution), while also extracting the full surplus.

3 Introducing Competition

One criticism of the model above is that retailers may not accept zero profit, due to the presence of other suppliers. In this section, we study how the conclusions above change in the presence of competition.

We model competition by assuming that the retailer has an outside option of value $\Pi \in [0, \Pi^*]$, with $\Pi = 0$ representing the case of a monopolist supplier, and $\Pi = \Pi^*$ representing the case of perfect competition among suppliers. We define $C = \Pi/\Pi^* \in [0, 1]$ to be the *level of competition*.⁵

As before, we can imagine that the supplier chooses r, q to solve the following problem:

$$\begin{aligned} \max_{r, q} \quad & (p(1 - F(q)) - c_h - c_m)q + r \left(qF(q) - \int_0^q F(x)dx \right) \\ \text{s.t.} \quad & (p - r)(qF(q) - \int_0^q F(x)dx) \geq \Pi, \end{aligned} \quad (7)$$

where the left side of the constraint represents expected retailer utility, as seen by substituting (3) into (2). As before for any fixed q , the supplier wishes to make r as large as possible while abiding by the profit constraint. For all q sufficiently large that the constraint is feasible, we note that the optimal choice of r is $r = p - \Pi/(qF(q) - \int_0^q F(x)dx)$. Substituting this into (7), we see that the supplier chooses q to maximize

$$(p - c_h - c_m)q - p \int_0^q F(x)dx.$$

The optimal choice of q is $q = q^* = F^{-1}(M)$, corresponding to a rebate of $r = p(1 - C)$ and a wholesale price of $w = p(1 - MC) - c_h$. We formalize this in Theorem 2.

Theorem 2. *In equilibrium with competition level C , we have $w + c_h = p(1 - MC)$, $r = p(1 - C)$, and $\frac{r}{w + c_h} = \frac{1 - C}{1 - MC}$. The quantity ordered by the retailer remains $q^* = F^{-1}(M)$.*

Remark 3. *There are several interesting observations. First, the optimal order quantity is independent of the level of competition C , and the optimal rebate level is independent of the margin M . Second, the “rebate generosity” $r/(w + c_h)$ (that is, the fraction of the supplier’s costs that can be recovered when inventory goes unsold) is decreasing in C and*

⁵One could imagine explicitly modeling different suppliers, each with different costs and offering different wholesale prices and rebates. So long as the retailer is committed to using a single supplier, in equilibrium the retailer should go with the “most efficient” supplier (the one that can offer the largest joint surplus between supplier and retailer): if the retailer chose a different supplier, then the most efficient supplier win the contract by offering more favorable terms. Furthermore, the profit of the retailer should be equal to the joint surplus offered by the “second most efficient” supplier: anything higher and the winning supplier is leaving something on the table. We capture this competition in reduced form with the utility guarantee Π .

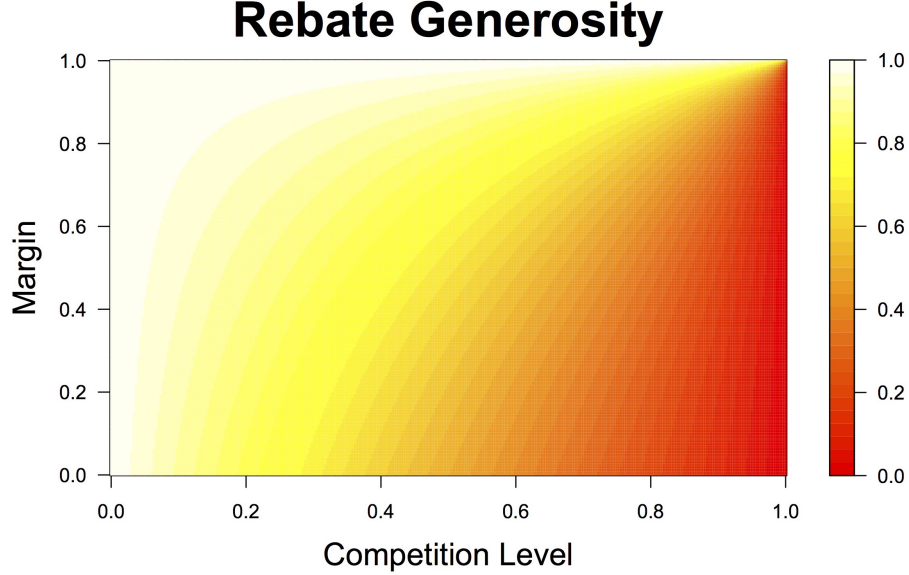


Figure 1: The generosity of rebates, as a function of the margin of the product and the level of competition in the marketplace. The supplier offers a near-complete rebate whenever the market is not competitive or the margin is high. In very competitive markets, rebates are small to non-existent.

increasing in M . In particular, if the margin is high or the level of competition is low, then the supplier offers a near-complete rebate. As the level of competition rises, both wholesale prices and rebate generosity fall, reaching $w = c_m$ and $r = 0$ in competitive markets. These trends are depicted in Figure 1.

While it may sound obvious that in a competitive market, wholesale prices fall to the marginal cost c_m , this is not immediate, as it is plausible that another supplier could “undercut” by offering a slightly higher price with a more generous rebate.

4 References

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